

CN-LAB-2

Q.1 what are the different protocols you observe at the following layers of the protocol stack?

A. Application layer

→ HTTP

TLS

DNS

B. Transport layer

→ TCP

UDP

C. Network layer

→ IPv4, IPv6

Q.2 What is the total amount of data being received from the following two cases.

A. when you access <http://neverssl.com/>

→ 3130 bytes received

B. When you access <https://www.cornell.edu>

→ 3854788 bytes (~3.85MB)

Q.3 Answer the following when you access the site <http://neversal.com/>

A. How many HTTP GET requests do you observe?
List down the GET requests.

→ GET /online

GET /online/

GET /favicon.ico

B. What version of HTTP the server is running?

→ HTTP/1.1

C. For each of the HTTP GET requests, find out:

(i) The total number of TCP segments being received

Get Request	TCP Segments received
1) (/online)	1
2) (/online/)	2
3) (/favicon.ico)	1

(ii) The total amount of data being received in the corresponding HTTP response message.

→ Get request

1) /online

2) /online/

3) /favicon.ico

Data is HTTP response

591 bytes

✓ 1519 bytes

416 bytes

D. Identify TCP three ways handshake used to establish the connection with the `http://neverssl.com`. Specify SYN, SYN-ACK, ACK packet and write the sequence number of each of the three packets.

→ Packet src - Dest

Seq No | Ack No

SYN client: 56670 → neverssl.com: 80 189603450

SYN-ACK neverssl.com: 80 → client: 56670 3023087423 189603451

ACK client: 56670 → neverssl.com: 80 189603451 3023087424